

Correlation (Pearson, Spearman)

Statistical Methods Dossier with Subject-Specific Examples

1. Method Description

Pearson correlation (r) measures the direction and strength of the linear relationship between two metric variables.

Range: -1 (perfect negative) to +1 (perfect positive). 0 = no linear relationship.

Spearman's rho: Rank-based correlation, robust to outliers and non-linearity.

Coefficient of determination: r^2 = proportion of shared variance.

A correlation of $r = 0.5$ means 25% shared variance ($r^2 = 0.25$).

2. Subject-Specific Examples

Social Sciences: Is income correlated with life satisfaction? $r = 0.32$, $p < .001$.

Economics: Are advertising spend and sales correlated? $r = 0.67$ (strong positive).

Human Sciences: Is therapy session count correlated with symptom reduction? $r = 0.45$.

Humanities: Are sentence length and word length correlated in literary texts? Spearman's rho.

3. Important Concepts and Assumptions

Pearson: linearity, normality (for inference), no outliers, homoscedasticity (for inference).

Spearman: Only requires monotonic relationship. No distributional assumptions.

Correlation does NOT imply causation: Confounders, reverse causality, and spurious correlations are possible.

Restriction of range: Limited variability reduces correlation magnitude.

4. Interpretation and practice

Report: 'A Pearson correlation showed a significant positive relationship between X and Y, $r(98) = .45$, $p < .001$, 95% CI [.28, .59].'

Effect size benchmarks: $|r| = .10$ (small), $.30$ (medium), $.50$ (large) per Cohen.

Always report confidence interval for r .

Partial correlation: Correlation between two variables after controlling for a third.

5. Code Examples

R: `cor.test(d$x, d$y, method = 'pearson');` `cor.test(d$x, d$y, method = 'spearman')`

Python: `from scipy import stats; stats.pearsonr(x, y); stats.spearmanr(x, y)`

SPSS: `CORRELATIONS /VARIABLES=x y /PRINT=TWOTAIL NOSIG /STATISTICS DESCRIPTIVES.`

R (partial): `library(ppcor); pcor.test(d$x, d$y, d$z)`